In [1]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt #Visualizing The data
%matplotlib inline
import seaborn as sns
```

In [11]:

```
df = pd.read_csv(r"C:\Users\sahil\OneDrive\Documents\diwali_data.csv", encoding= 'unicode_escape')
#to avoid error, using unicode_escape
```

In [12]:

```
df.head()
```

Out[12]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_Category	Orders	Amount	Status	unnamed1
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	Auto	1	23952.0	NaN	NaN
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	Auto	3	23934.0	NaN	NaN
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	Auto	3	23924.0	NaN	NaN
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	Auto	2	23912.0	NaN	NaN
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	Auto	2	23877.0	NaN	NaN

In [13]:

```
df.shape
```

Out[13]:

```
(11251, 15)
```

In [15]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   User_ID           11251 non-null  int64  
 1   Cust_name         11251 non-null  object 
 2   Product_ID        11251 non-null  object 
 3   Gender            11251 non-null  object 
 4   Age Group         11251 non-null  object 
 5   Age               11251 non-null  int64  
 6   Marital_Status    11251 non-null  int64  
 7   State             11251 non-null  object 
 8   Zone              11251 non-null  object 
 9   Occupation        11251 non-null  object 
 10  Product_Category  11251 non-null  object 
 11  Orders            11251 non-null  int64  
 12  Amount            11239 non-null  float64
 13  Status            0 non-null      float64
 14  unnamed1          0 non-null      float64
dtypes: float64(3), int64(4), object(8)
memory usage: 1.3+ MB
```

In [16]:

```
df.drop(['Status', 'unnamed1'], axis=1, inplace=True)
#drops unrelated or blank colums
```

In [17]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   User_ID           11251 non-null  int64  
 1   Cust_name         11251 non-null  object 
 2   Product_ID        11251 non-null  object 
 3   Gender            11251 non-null  object 
 4   Age Group         11251 non-null  object 
 5   Age               11251 non-null  int64  
 6   Marital_Status    11251 non-null  int64  
 7   State             11251 non-null  object 
 8   Zone              11251 non-null  object 
 9   Occupation        11251 non-null  object 
 10  Product_Category  11251 non-null  object 
 11  Orders            11251 non-null  int64  
 12  Amount            11239 non-null  float64
dtypes: float64(1), int64(4), object(8)
memory usage: 1.1+ MB
```

In [20]:

```
pd.isnull(df)
#checking null values
```

Out[20]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_Category	Orders	Amount
0	False	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...
11246	False	False	False	False	False	False	False	False	False	False	False	False	False
11247	False	False	False	False	False	False	False	False	False	False	False	False	False
11248	False	False	False	False	False	False	False	False	False	False	False	False	False
11249	False	False	False	False	False	False	False	False	False	False	False	False	False
11250	False	False	False	False	False	False	False	False	False	False	False	False	False

11251 rows × 13 columns

In [21]:

```
pd.isnull(df).sum()
```

Out[21]:

```
User_ID              0
Cust_name            0
Product_ID           0
Gender               0
Age Group            0
Age                  0
Marital_Status       0
State                0
Zone                 0
Occupation           0
Product_Category     0
Orders               0
Amount              12
dtype: int64
```

In [22]:

```
df.shape
```

Out[22]:

```
(11251, 13)
```

In [23]:

```
df.dropna(inplace=True)
# droping null values
```

In [24]:

```
df.shape
```

Out[24]:

```
(11239, 13)
```

In [26]:

```
df['Amount'] = df['Amount'].astype('int')
#chinging data  type to float vale to int
```

In [27]:

```
df['Amount'].dtypes
```

Out[27]:

```
dtype('int64')
```

In [30]:

```
df.columns
```

Out[30]:

```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='object')
```

In [32]:

```
df.rename (columns= {'Marital_Status' : 'Shadi'})
```

Out[32]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Shadi	State	Zone	Occupation	Product_Category	Orders	Amount
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	Auto	1	23952
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	Auto	3	23934
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	Auto	3	23924
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	Auto	2	23912
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	Auto	2	23877
...	...	...	...	...	...	...	...	...	...	...	...	...	...
11246	1000695	Manning	P00296942	M	18-25	19	1	Maharashtra	Western	Chemical	Office	4	370
11247	1004089	Reichenbach	P00171342	M	26-35	33	0	Haryana	Northern	Healthcare	Veterinary	3	367
11248	1001209	Oshin	P00201342	F	36-45	40	0	Madhya Pradesh	Central	Textile	Office	4	213
11249	1004023	Noonan	P00059442	M	36-45	37	0	Karnataka	Southern	Agriculture	Office	3	206
11250	1002744	Brumley	P00281742	F	18-25	19	0	Maharashtra	Western	Healthcare	Office	3	188

11239 rows × 13 columns

In [34]:

```
df.describe()
# description of the data in data frame ex: count, mean, std, etc
```

Out[34]:

	User_ID	Age	Marital_Status	Orders	Amount
count	1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
mean	1.003004e+06	35.410357	0.420055	2.489634	9453.610553
std	1.716039e+03	12.753866	0.493589	1.114967	5222.355168
min	1.000001e+06	12.000000	0.000000	1.000000	188.000000
25%	1.001492e+06	27.000000	0.000000	2.000000	5443.000000
50%	1.003064e+06	33.000000	0.000000	2.000000	8109.000000
75%	1.004426e+06	43.000000	1.000000	3.000000	12675.000000
max	1.006040e+06	92.000000	1.000000	4.000000	23952.000000

In [36]:

```
df[['Age', 'Orders', 'Amount']].describe()
```

Out[36]:

	Age	Orders	Amount
count	11239.000000	11239.000000	11239.000000
mean	35.410357	2.489634	9453.610553
std	12.753866	1.114967	5222.355168
min	12.000000	1.000000	188.000000
25%	27.000000	2.000000	5443.000000
50%	33.000000	2.000000	8109.000000
75%	43.000000	3.000000	12675.000000
max	92.000000	4.000000	23952.000000

In [37]:

```
#Exploratory Data Analysis
```

In [38]:

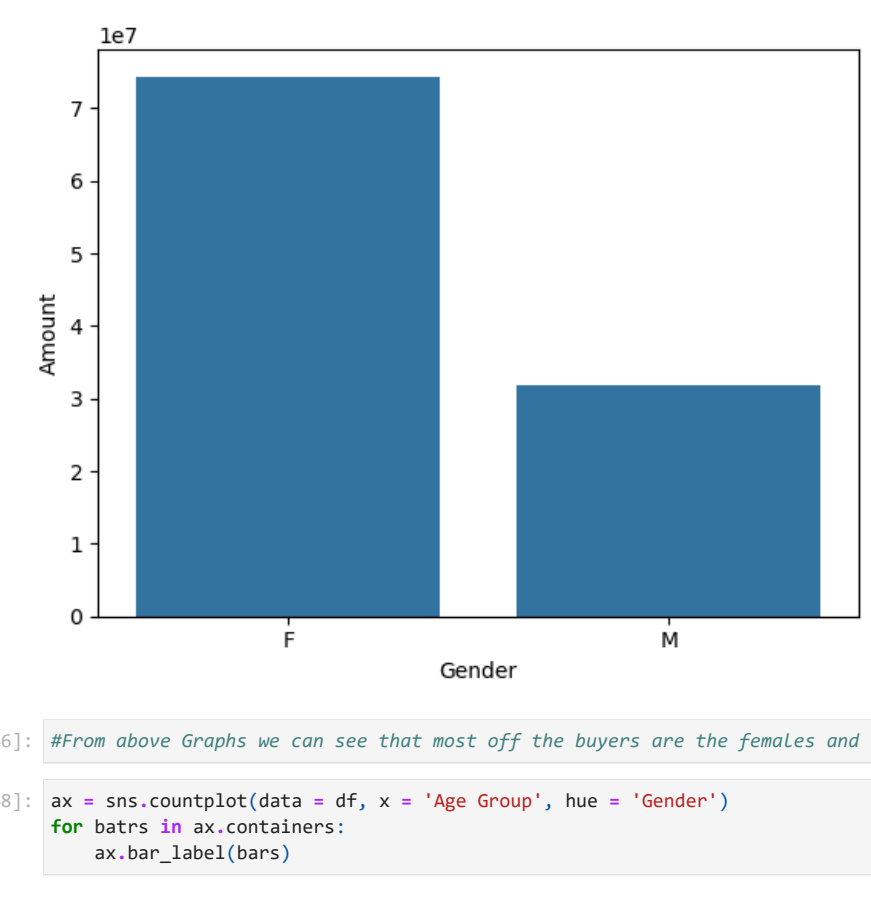
```
df.columns
```

Out[38]:

```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='object')
```

In [43]:

```
ax = sns.countplot(x = 'Gender',data = df)
for bars in ax.containers:
    ax.bar_label(bars)
```



In [44]:

```
df.groupby(['Gender'], as_index=False) ['Amount'].sum().sort_values(by = 'Amount', ascending=False)
```

Out[44]:

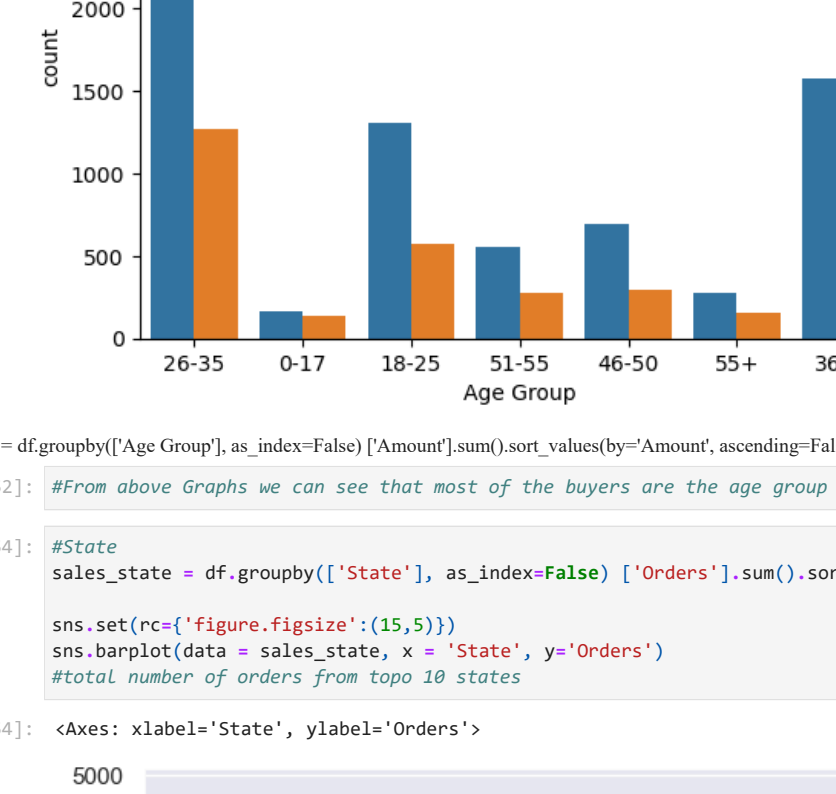
	Gender	Amount
0	F	74335853
1	M	31913276

In [45]:

```
sales_gen = df.groupby(['Gender'], as_index=False) ['Amount'].sum().sort_values(by = 'Amount', ascending=False)
sns.barplot(x = 'Gender', y = 'Amount', data = sales_gen)
```

Out[45]:

```
<Axes: xlabel='Gender', ylabel='Amount'>
```

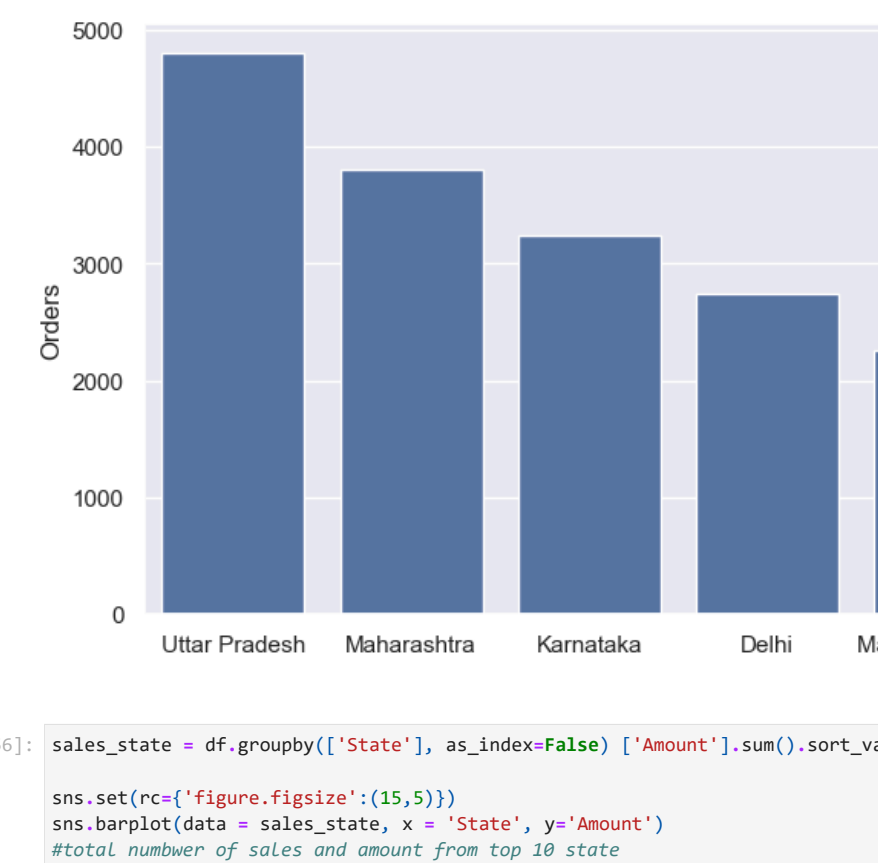


In [46]:

```
#From above Graphs we can see that most off the buyers are the females and even the puchasing power of the females are grotter than male
```

In [48]:

```
ax = sns.countplot(data = df, x = 'Age Group', hue = 'Gender')
for batrs in ax.containers:
    ax.bar_label(bars)
```



sales_age = df.groupby(['Age Group'], as_index=False) ['Amount'].sum().sort_values(by='Amount', ascending=False) sns.barplot(x = 'Age Group', y = 'Amount', data = sales_age) #total Amount Vs Age group

In [52]:

```
#From above Graphs we can see that most of the buyers are the age group between 26-35yrs female
```

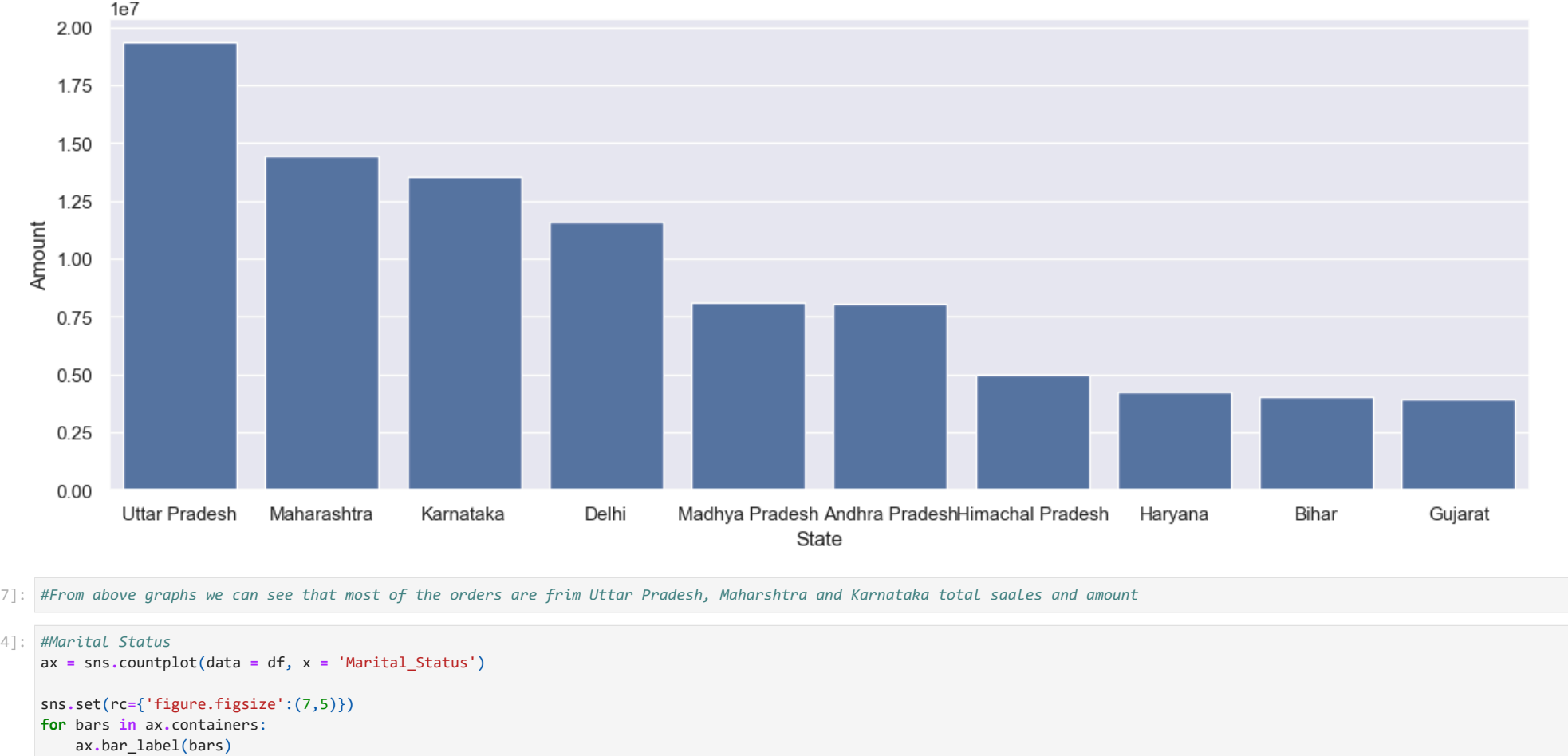
In [54]:

```
#State
sales_state = df.groupby(['State'], as_index=False) ['Orders'].sum().sort_values(by='Orders', ascending=False).head(10)

sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(data = sales_state, x = 'State', y='Orders')
#total number of orders from topo 10 states
```

Out[54]:

```
<Axes: xlabel='State', ylabel='Orders'>
```



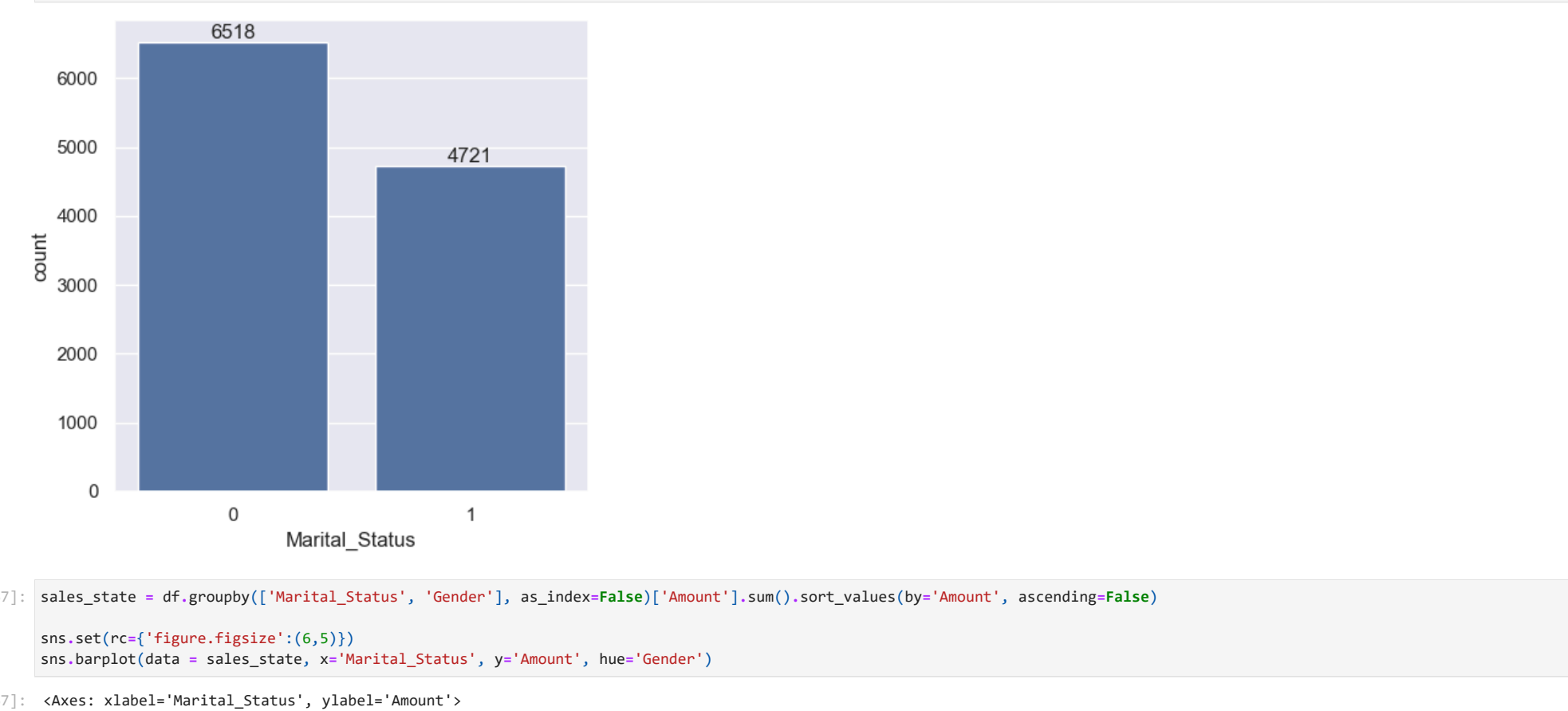
In [56]:

```
sales_state = df.groupby(['State'], as_index=False) ['Amount'].sum().sort_values(by='Amount', ascending=False).head(10)

sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(data = sales_state, x = 'State', y='Amount')
#total numbwer of sales and amount from top 10 state
```

Out[56]:

```
<Axes: xlabel='State', ylabel='Amount'>
```



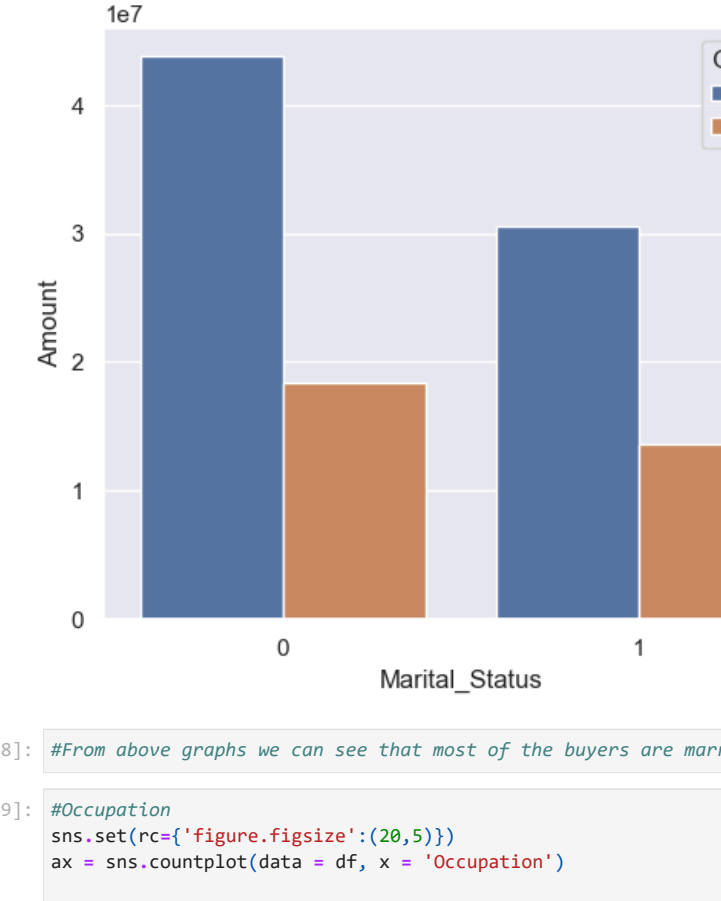
In [57]:

```
#From above graphs we can see that most of the orders are frim Uttar Pradesh, Maharshtra and Karnataka total saales and amount
```

In [64]:

```
#Marital Status
ax = sns.countplot(data = df, x = 'Marital_Status')

sns.set(rc={'figure.figsize':(7,5)})
for bars in ax.containers:
    ax.bar_label(bars)
```



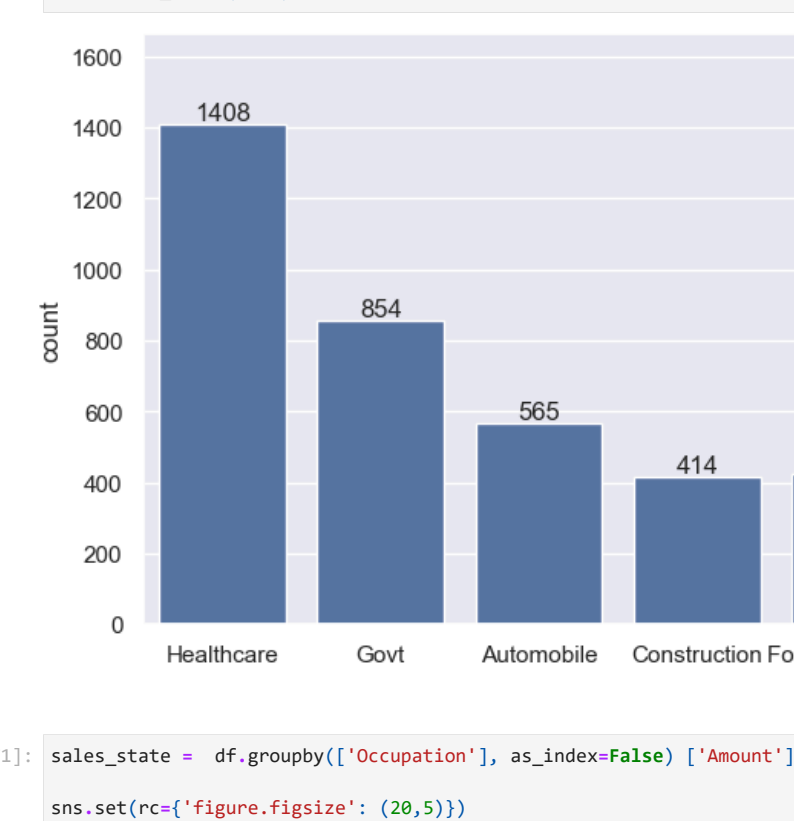
In [67]:

```
sales_state = df.groupby(['Marital_Status', 'Gender'], as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False)

sns.set(rc={'figure.figsize':(6,5)})
sns.barplot(data = sales_state, x='Marital_Status', y='Amount', hue='Gender')
```

Out[67]:

```
<Axes: xlabel='Marital_Status', ylabel='Amount'>
```



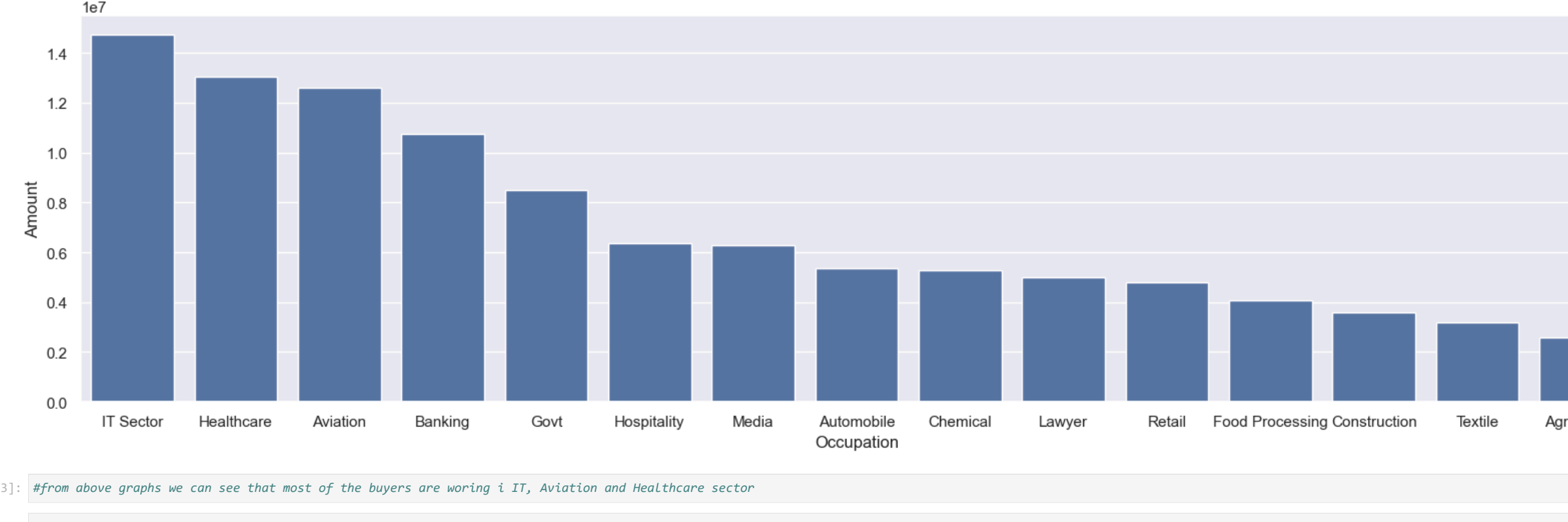
In [68]:

```
#From above graphs we can see that most of the buyers are married women and they have high purchasing power
```

In [69]:

```
#Occupation
sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot(data = df, x = 'Occupation')

for bars in ax.containers:
    ax.bar_label(bars)
```



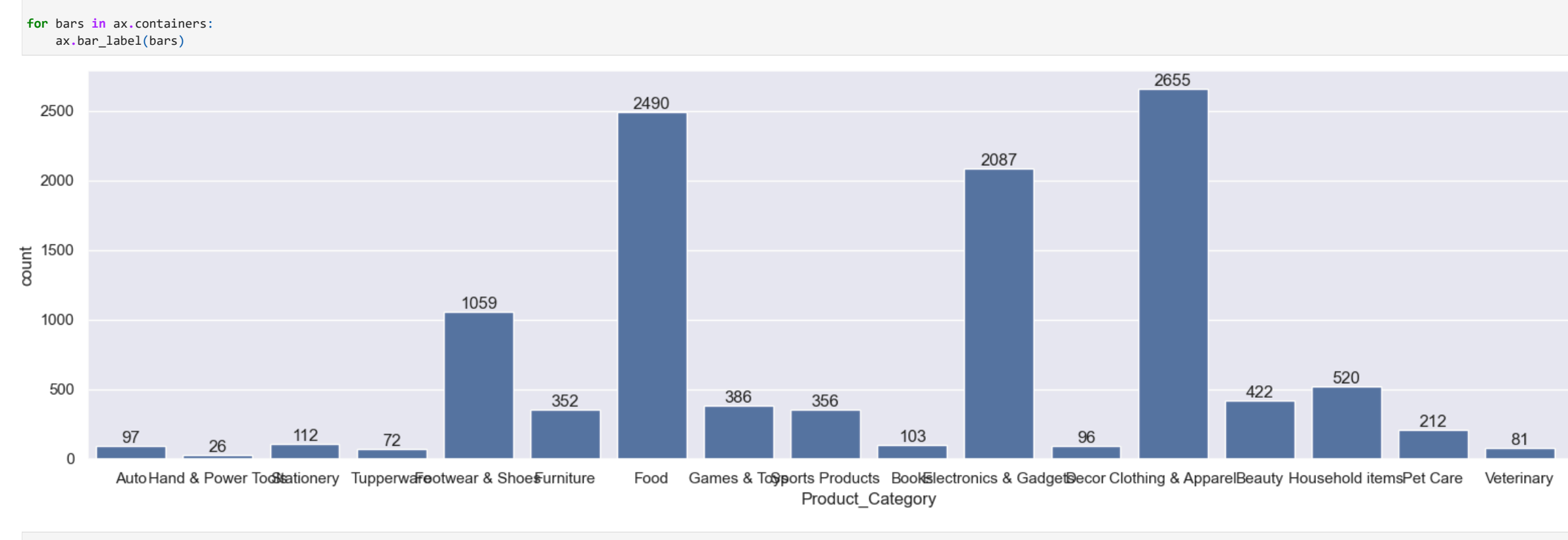
In [71]:

```
sales_state =  df.groupby(['Occupation'], as_index=False) ['Amount'].sum().sort_values(by='Amount', ascending=False)

sns.set(rc={'figure.figsize': (20,5)})
sns.barplot(data = sales_state, x='Occupation', y='Amount')
```

Out[71]:

```
<Axes: xlabel='Occupation', ylabel='Amount'>
```



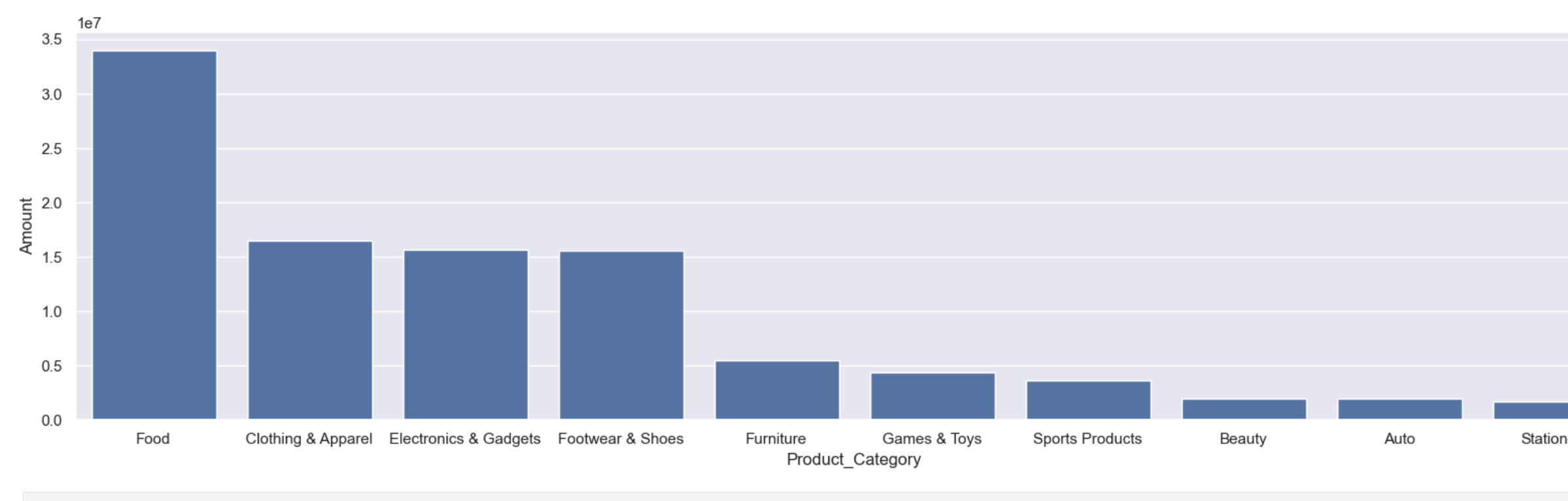
In [73]:

```
#from above graphs we can see that most of the buyers are woring i IT, Aviation and Healthcare sector
```

In [74]:

```
#Product Category
sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot( data = df, x = 'Product_Category')

for bars in ax.containers:
    ax.bar_label(bars)
```



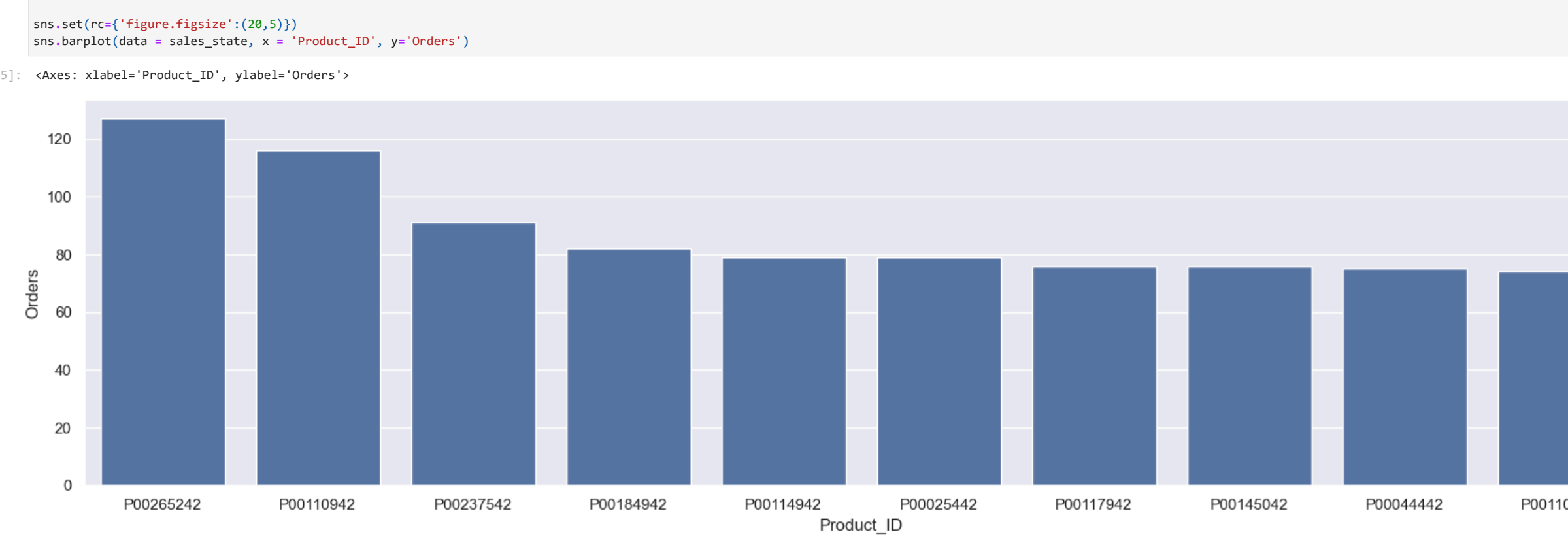
In [77]:

```
sales_state = df.groupby(['Product_Category'], as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False).head(10)

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_Category', y='Amount')
```

Out[77]:

```
<Axes: xlabel='Product_Category', ylabel='Amount'>
```



In [76]:

```
 #from above graphs we can see that most of the sold products are from food, footwear and electroinc
```

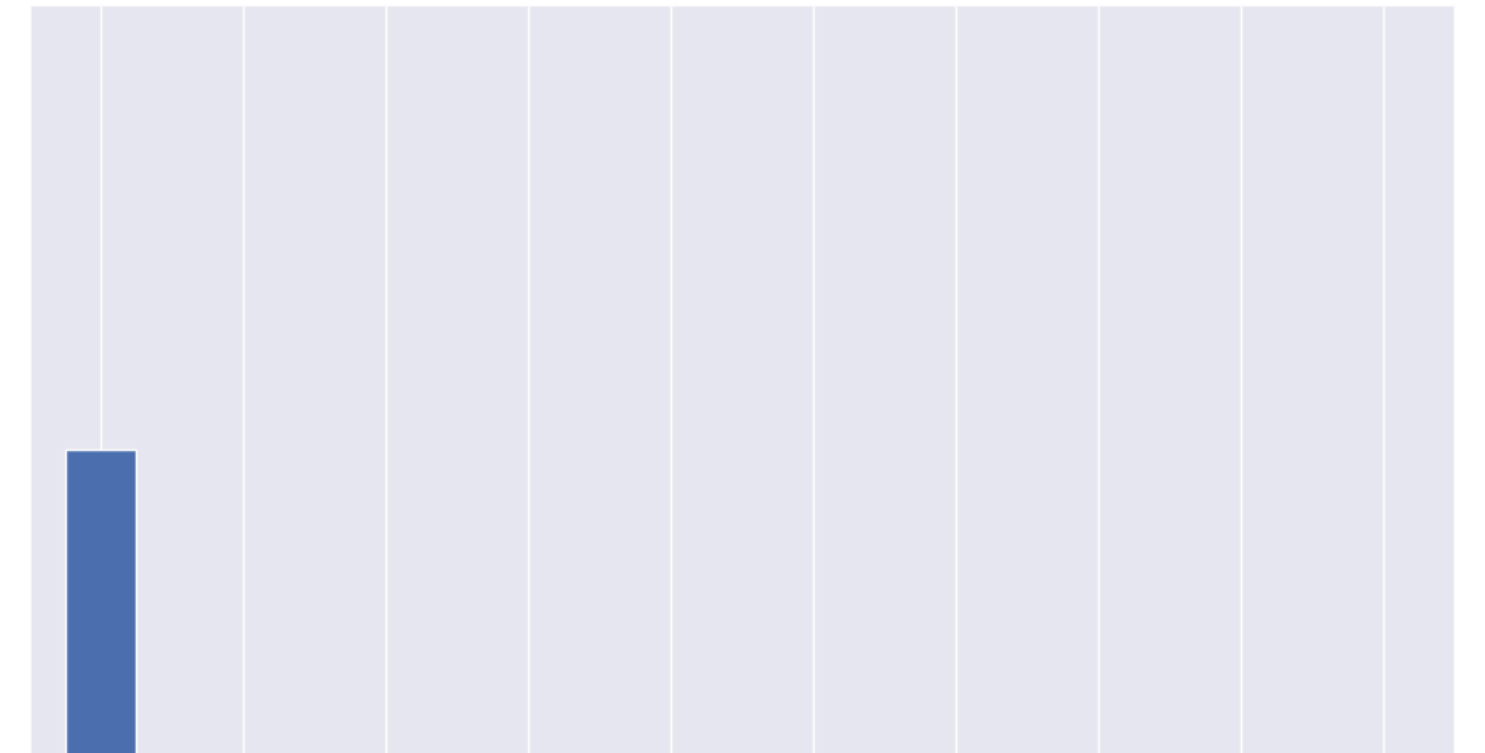
In [85]:

```
sales_state = df.groupby(['Product_ID'], as_index=False)['Orders'].sum().sort_values(by='Orders', ascending=False).head(10)

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_ID', y='Orders')
```

Out[85]:

```
<Axes: xlabel='Product_ID', ylabel='Orders'>
```

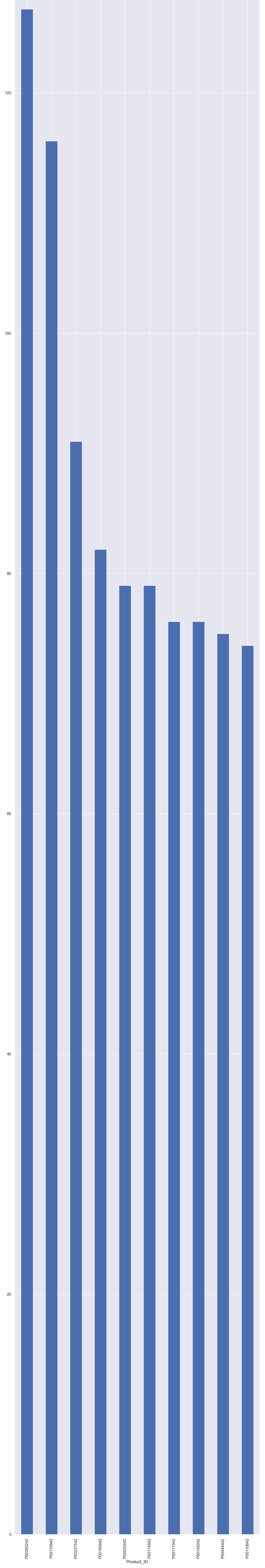


In [86]:

```
fig1, ax1 = plt.subplots(figsize=(12,79))
df.groupby('Product_ID')['Orders'].sum().nlargest(10).sort_values(ascending=False).plot(kind='bar')
```

Out[86]:

```
<Axes: xlabel='Product_ID'>
```



In [ ]:

In [ ]:

```
In [ ]:
```

```
In [ ]:
```